



JAY KRISSELL
Electrical Engineer

CONTACT

jaykrisell@outlook.com
(501) 269-5750
Fayetteville, AR

LINKS

[LinkedIn Profile](#)
[GitHub — SpaceCadetJ](#)
[Portfolio Website](#)

INTERESTS

- Embedded Control Systems
- Reinforcement Learning
- Motor Control & FOC
- Motion Planning
- PCB Design & Fabrication
- Robotics Commercialization

REFERENCES

Dr. Tuan Dang
Asst. Professor, EECS
University of Arkansas

Mike Kieffer
VP, Keenfinity
612-237-6522

SFC Brandon Dostert
Readiness NCO, 1-185th C-Co
479-653-6110

WORK EXPERIENCE

Cognitive Robotics Lab — University of Arkansas — Fayetteville, AR **Graduate Research Assistant**

Jan 2025 – Present

- Conducting reinforcement learning research under Dr. Tuan Dang, comparing PPO and AWR policy optimization for motion imitation in humanoid robots.
- Deploying simulation environments in NVIDIA IsaacGym and MimicKit for physics-based motion capture replay and control policy training.
- Developing custom reward functions and training pipelines for robotic actuator control using PyTorch-based RL frameworks.

U.S. Army National Guard — Camp Robinson, AR **Aviation Electrician (15F) — Sergeant**

Dec 2019 – Dec 2025

- Led avionics maintenance on UH-60 and UH-72 airframes: sensors, communications, imaging systems, IVHMS, and flight instruments.
- Supervised maintenance teams and maintained readiness documentation as non-commissioned officer.
- Completed 7-month AIT at JBLE Fort Eustis covering UH-60 and CH-47 avionics. Graduated with aviation maintainer certification.

Bosch (Electro-Voice R&D) — Burnsville, MN **Quality Engineer Co-Op**

Jul 2023 – Dec 2023

- Designed and validated prototype electronics and automated reliability tests for professional loudspeakers.
- Conducted DSP, endurance, stress, and frequency response testing in anechoic chambers using laser vibrometry.
- Coordinated with customers on purchase orders and quality constraints for developing and competitive products.

EDUCATION

University of Arkansas — 2025 – Present

M.S. Electrical Engineering (In Progress)

Cognitive Robotics Lab · Reinforcement learning, robotic control, embedded intelligence

University of Arkansas — 2021 – 2025

B.S. Electrical Engineering

Circuit design, robotics, control systems

JBLE Fort Eustis — U.S. Army — 2020 – 2021

Helicopter Avionics AIT (15F)

7-month advanced training · Aviation maintainer certification

University of Arkansas — Little Rock — 2018 – 2020

Associate of Arts

Concurrent enrollment · Graduated high school top 5%

CORE SKILLS

- Python · PyTorch · RL
- C/C++ · Embedded
- MATLAB · Simulink
- ROS2 · Gazebo · IsaacGym
- KiCad · Cadence Allegro
- STM32 · ARM Cortex-M
- FOC / BLDC Motor Control
- PCB Design (6-layer)
- CAN Bus · LoRa · SPI/I2C
- SMD Soldering · Rework
- Git · Linux · Docker
- Circuit Troubleshooting

COURSEWORK

- Cognitive Robotics
- Robot Localization (KF/EKF)
- Control Systems
- Electromagnetics
- Energy Systems
- Signals & Systems

CERTIFICATIONS

- FAA Aviation Maintainer
- Army 15F MOS Qualified
- Secret Security Clearance

LEADERSHIP

Six years as an Army NCO leading avionics maintenance teams under strict safety and readiness standards. Cross-functional coordination at Bosch R&D. Founded and operating an early-stage deep-tech robotics venture.

PROJECTS & VENTURES

Kinetic Intelligence Labs — TITAN FOC v8.2

Founder & Lead Engineer

Oct 2024 – Present

- Founded a robotics hardware venture developing custom field-oriented control (FOC) motor controllers and RL-based training systems for robotic actuators. Targeting the gap between expensive industrial drives and hobbyist-grade boards.
- Designed the TITAN FOC v8.2: a 50.25mm circular 6-layer PCB built around the STM32G474CEU6 MCU and DRV8353RS three-phase gate driver, with six TPW1R306PL MOSFETs, AS5047P magnetic encoder, and TCAN1051 CAN transceiver. 88 components, \$32.39 BOM — fabricated by PCBWay with ENIG finish and 2oz outer copper.
- Architected a 16-sheet hierarchical KiCad schematic following the Amulet/KDT professional template pattern. Developed integrated AI-assisted layout review tooling via a custom MCP server connected to Claude Desktop.
- Exploring university IP commercialization pathways and building go-to-market strategy for the robotic joint actuator and light EV drivetrain segments.

PPO vs. AWR Motion Imitation Research

Cognitive Robotics Lab · 2025 – Present

- Benchmarking Proximal Policy Optimization against Advantage-Weighted Regression for humanoid whole-body motion imitation using NVIDIA IsaacGym physics simulation with MimicKit motion-capture replay.
- Built the full simulation pipeline: WSL2 with CUDA 12 GPU passthrough, conda-managed environments, and custom reward shaping for joint-level tracking accuracy. Research targets comparing sample efficiency, policy robustness, and sim-to-real transfer potential.

LoRa Autonomous RC Boat Fleet

Personal Project · 2024

- Designed and built custom STM32F446-based control boards for a fleet of autonomous surface vehicles with LoRa mesh networking for beyond-line-of-sight communication, u-blox GPS waypoint navigation, CAN bus sensor fusion, and a centralized ground station for real-time telemetry and multi-agent formation control.

Humanoid Robot Simulation

University of Arkansas · 2024

- Built a 24-degree-of-freedom humanoid simulation in ROS2 Humble and Gazebo with a full URDF model, TF transform tree, C++ and Python controller nodes, ros2_control joint trajectory interface, and scripted mission replay for gait cycle analysis.

Robotic Arm Simulator & GUI

University of Arkansas · 2024

- Developed an adjustable-DOF robotic arm simulator with a Tkinter GUI supporting full range-of-motion remote control, real-time inverse and forward kinematics computation, end-effector position tracking, collision detection, and 3D visualization.